

Advanced Engineering Mathematics

Integral Transform Methods

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1 Integral Transform Methods

Key Concept

An integral transform converts a function from one domain into another domain by integrating it against a kernel:

$$F(\lambda) = \int_a^b K(\lambda, t) f(t) dt.$$

Here, $f(t)$ is the original function, $K(\lambda, t)$ is the kernel, and $F(\lambda)$ is the transformed function.

Integral transform methods are powerful because they replace difficult operations in the original domain with simpler operations in the transform domain. In engineering mathematics, the most common use is to convert differential equations, convolution problems, and signal-processing problems into algebraic or spectral forms.

The central idea is:

difficult problem in t $\longrightarrow$ simpler problem in transform variable $\longrightarrow$ inverse transform.

Engineering Note

In engineering, transform methods are used in circuit analysis, control systems, signal processing, vibration analysis, communication systems, image processing, and numerical solution of differential equations.

General Workflow

$$f(t) \xrightarrow{\text{transform}} F(\lambda) \xrightarrow{\text{solve or analyze}} \text{simpler expression} \xrightarrow{\text{inverse transform}} f(t)$$

Key Concept

The four transform methods in this lecture are:

- Laplace Transform for causal systems and initial-value problems,
- Fourier Transform for frequency analysis of aperiodic signals,
- Hilbert Transform for phase, analytic signals, and spectral relationships,
- Z Transform for discrete-time sequences and digital systems.

Connection to Previous Topics

Fourier series expanded periodic functions as sums of sine and cosine waves. Integral transforms extend this idea to broader classes of functions.

Fourier series handles periodic signals, while the Fourier transform handles aperiodic signals. The Laplace transform extends this further by adding exponential damping through e^{-st} , making it especially useful for functions defined for $t > 0$.

Engineering Note

A useful way to remember the difference is:

Fourier Series : periodic signal $\rightarrow$ discrete frequencies,

Fourier Transform : aperiodic signal $\rightarrow$ continuous frequencies,

Laplace Transform : causal signal $\rightarrow$ complex-frequency model,

Z Transform : discrete signal $\rightarrow$ complex discrete-frequency model.

Example

Example: Why transforms are useful

Suppose an engineering system is modeled by

$$y''(t) + 3y'(t) + 2y(t) = x(t).$$

In the time domain, this is a differential equation. After applying a transform, derivatives become algebraic factors. For example, under the Laplace transform,

$$L[y'(t)] = sY(s) - y(0),$$

and

$$L[y''(t)] = s^2Y(s) - sy(0) - y'(0).$$

Thus, the differential equation becomes an algebraic equation in $Y(s)$:

$$(s^2Y(s) - sy(0) - y'(0)) + 3(sY(s) - y(0)) + 2Y(s) = X(s).$$

Solving for $Y(s)$ is often easier than solving directly for $y(t)$.

Common Mistakes

Key Concept

Common mistakes in transform methods include:

1. Forgetting the domain of the transform variable.
2. Ignoring convergence conditions.
3. Applying inverse transforms without checking the form.
4. Confusing continuous-time transforms with discrete-time transforms.
5. Treating transform tables as formulas without understanding assumptions.

Engineering Note

The transform is not just a computational shortcut. It changes the viewpoint of a problem. In engineering, this is often the difference between analyzing a signal in time and analyzing the same signal in frequency, stability, or pole-zero form.

1.1 Laplace Transform

Motivation and Intuition

The Laplace transform is designed for functions that begin at $t = 0$ and are usually taken to be zero for $t < 0$. This makes it natural for engineering systems that start operating at a particular time, such as circuits after a switch is closed, mechanical systems after a force is applied, or control systems after an input signal begins.

The main advantage is that it converts differentiation into multiplication by s , while also carrying initial conditions into the transformed equation.

Key Concept

For a function $f(t)$ such that $f(t) = 0$ for $t < 0$, the Laplace transform is defined by

$$L[f(t)] = F(s) = \int_0^{\infty} f(t)e^{-st} dt.$$

The inverse transform is written as

$$L^{-1}[F(s)] = f(t).$$

The factor e^{-st} acts like a damping or weighting factor. If $f(t)$ grows too quickly, the integral may not converge. Therefore, functions are usually required to be piecewise continuous

and of exponential order.

Key Concept

A function $f(t)$ is of exponential order if there exist constants M and k such that

$$|f(t)| \leq Me^{kt}, \quad t > 0.$$

Laplace Transforms of Common Functions

Key Concept

The following table summarizes commonly used Laplace transforms. These are especially useful when solving initial-value problems and circuit/system models.

Function $f(t)$, $t \geq 0$	Laplace Transform $F(s) = L[f(t)]$
1	$\frac{1}{s}$
e^{-at}	$\frac{1}{s+a}$
e^{at}	$\frac{1}{s-a}$
t	$\frac{1}{s^2}$
t^n , $n = 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}$
$\sin(at)$	$\frac{a}{s^2 + a^2}$
$\cos(at)$	$\frac{s}{s^2 + a^2}$
$\sinh(at)$	$\frac{a}{s^2 - a^2}$
$\cosh(at)$	$\frac{s}{s^2 - a^2}$
$H(t-a)$	$\frac{e^{-as}}{s}$
$\delta(t-a)$	e^{-as}

Engineering Note

The Laplace transform table is most useful when the function is causal, meaning the system response begins at $t = 0$. In engineering, this matches switched circuits, forced mechanical systems, and control-system inputs.

Key Properties

Key Concept

Linearity

$$L[c_1 f(t) + c_2 g(t)] = c_1 L[f(t)] + c_2 L[g(t)].$$

Key Concept

Derivative Property If $F(s) = L[f(t)]$, then

$$L[f'(t)] = sF(s) - f(0).$$

For the second derivative,

$$L[f''(t)] = s^2 F(s) - sf(0) - f'(0).$$

This property is the reason Laplace transforms are especially effective for initial-value problems.

Example

Example 1: Find the Laplace transform of $f(t) = 3e^{2t} - 4\sin(5t)$.

Using linearity,

$$L[3e^{2t} - 4\sin(5t)] = 3L[e^{2t}] - 4L[\sin(5t)].$$

From the basic transform formulas,

$$L[e^{2t}] = \frac{1}{s-2},$$

and

$$L[\sin(5t)] = \frac{5}{s^2 + 25}.$$

Therefore,

$$F(s) = \frac{3}{s-2} - \frac{20}{s^2 + 25}.$$

Example

Example 2: Find $L^{-1} \left[\frac{s+2}{s^2+1} \right]$.

Separate the expression:

$$\frac{s+2}{s^2+1} = \frac{s}{s^2+1} + \frac{2}{s^2+1}.$$

Using the table,

$$L[\cos t] = \frac{s}{s^2+1},$$

and

$$L[\sin t] = \frac{1}{s^2+1}.$$

Thus,

$$L^{-1} \left[\frac{s}{s^2+1} \right] = \cos t,$$

and

$$L^{-1} \left[\frac{2}{s^2+1} \right] = 2 \sin t.$$

Therefore,

$$f(t) = \cos t + 2 \sin t.$$

Example

Example 3: Solve the initial-value problem

$$y'(t) + 2y(t) = e^{-t}, \quad y(0) = 3.$$

Apply the Laplace transform:

$$L[y'(t)] + 2L[y(t)] = L[e^{-t}].$$

Using

$$L[y'(t)] = sY(s) - y(0),$$

we get

$$sY(s) - 3 + 2Y(s) = \frac{1}{s+1}.$$

Combine the $Y(s)$ terms:

$$(s+2)Y(s) - 3 = \frac{1}{s+1}.$$

Solve for $Y(s)$:

$$(s+2)Y(s) = 3 + \frac{1}{s+1}.$$

$$Y(s) = \frac{3}{s+2} + \frac{1}{(s+1)(s+2)}.$$

Use partial fractions:

$$\frac{1}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}.$$

Multiplying by $(s+1)(s+2)$ gives

$$1 = A(s+2) + B(s+1).$$

Set $s = -1$:

$$1 = A(1),$$

so

$$A = 1.$$

Set $s = -2$:

$$1 = B(-1),$$

so

$$B = -1.$$

Therefore,

$$Y(s) = \frac{3}{s+2} + \frac{1}{s+1} - \frac{1}{s+2}.$$

$$Y(s) = \frac{1}{s+1} + \frac{2}{s+2}.$$

Taking the inverse Laplace transform,

$$y(t) = e^{-t} + 2e^{-2t}.$$

Engineering Insight

Engineering Note

In circuit analysis, capacitors and inductors produce differential equations. The Laplace transform converts these equations into algebraic equations involving impedances. For example, differentiation in time corresponds to multiplication by s , allowing engineers to analyze transient and steady-state behavior in one framework.

Common Mistakes

Key Concept

Common mistakes in the Laplace transform include:

1. Forgetting initial conditions in $L[y'(t)]$ and $L[y''(t)]$.
2. Writing $L[e^{at}] = \frac{1}{s+a}$ instead of $\frac{1}{s-a}$.
3. Applying transform formulas without checking convergence.
4. Forgetting to decompose rational functions before inversion.

1.2 Fourier Transform

Motivation and Intuition

Fourier series represent periodic functions as sums of sine and cosine functions. However, many engineering signals are not periodic. The Fourier transform extends the idea of Fourier series to aperiodic functions by allowing a continuous range of frequencies.

Instead of writing a signal as a discrete sum of harmonics, the Fourier transform represents the signal as a continuous superposition of complex exponentials.

Key Concept

The Fourier transform of $f(t)$ is

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt.$$

The inverse Fourier transform is

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega)e^{i\omega t} d\omega.$$

The variable ω represents angular frequency. The function $F(\omega)$ describes how much of each frequency is present in $f(t)$.

Physical Interpretation

Engineering Note

A useful analogy is that the Fourier transform acts like a prism. A prism separates white light into colors; the Fourier transform separates a signal into its frequency components. The inverse Fourier transform recombines those components to reconstruct the original signal.

The magnitude

$$|F(\omega)|$$

is called the amplitude spectrum, while the argument

$$\arg F(\omega)$$

is the phase spectrum.

Key Properties

Key Concept

Linearity

$$\mathcal{F}[c_1 f(t) + c_2 g(t)] = c_1 F(\omega) + c_2 G(\omega).$$

Key Concept

Time Shifting If

$$\mathcal{F}[f(t)] = F(\omega),$$

then

$$\mathcal{F}[f(t - \tau)] = e^{-i\omega\tau} F(\omega).$$

Key Concept

Scaling If k is a nonzero real constant, then

$$\mathcal{F}[f(kt)] = \frac{1}{|k|} F\left(\frac{\omega}{k}\right).$$

Key Concept

Differentiation If

$$\mathcal{F}[f(t)] = F(\omega),$$

then

$$\mathcal{F}[f'(t)] = i\omega F(\omega).$$

More generally,

$$\mathcal{F}[f^{(n)}(t)] = (i\omega)^n F(\omega).$$

Key Concept

Parseval's Equality

$$\int_{-\infty}^{\infty} |f(t)|^2 dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |F(\omega)|^2 d\omega.$$

This connects energy in the time domain with energy in the frequency domain.

Fourier Transforms of Common Functions

Key Concept

Using the convention

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt,$$

some commonly encountered Fourier transform pairs are:

Function $f(t)$	Fourier Transform $F(\omega)$
$e^{-at}H(t), a > 0$	$\frac{1}{a + i\omega}$
$e^{at}H(-t), a > 0$	$\frac{1}{a - i\omega}$
$te^{-at}H(t), a > 0$	$\frac{1}{(a + i\omega)^2}$
$t^n e^{-at}H(t), \text{Re}(a) > 0$	$\frac{n!}{(a + i\omega)^{n+1}}$
$e^{-a t }, a > 0$	$\frac{2a}{\omega^2 + a^2}$
$te^{-a t }, a > 0$	$\frac{-4a\omega i}{(\omega^2 + a^2)^2}$
$\frac{1}{1 + a^2 t^2}$	$\frac{\pi}{ a } e^{- \omega/a }$
$\begin{cases} 1, & t < a, \\ 0, & t > a \end{cases}$	$\frac{2 \sin(\omega a)}{\omega}$
$\frac{\sin(at)}{at}$	$\begin{cases} \frac{\pi}{a}, & \omega < a, \\ 0, & \omega > a \end{cases}$
$e^{-at^2}, a > 0$	$\sqrt{\frac{\pi}{a}} \exp\left(-\frac{\omega^2}{4a}\right)$

Key Concept

Some Fourier transforms require the Dirac delta function:

Function $f(t)$	Fourier Transform $F(\omega)$
1	$2\pi\delta(\omega)$
$e^{i\omega_0 t}$	$2\pi\delta(\omega - \omega_0)$
$\cos(\omega_0 t)$	$\pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$
$\sin(\omega_0 t)$	$-\pi i\delta(\omega - \omega_0) + \pi i\delta(\omega + \omega_0)$
$H(t)$	$\pi\delta(\omega) + \frac{1}{i\omega}$

Engineering Note

The Fourier transform table helps engineers quickly identify the frequency content of signals. For example, rectangular pulses transform into sinc-type functions, while pure sinusoids transform into impulses at their frequencies.

Worked Examples

Example

Example 1: Fourier transform of a rectangular pulse

Let

$$f(t) = \begin{cases} 1, & |t| < a, \\ 0, & |t| > a. \end{cases}$$

Using the definition,

$$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t} dt.$$

Since $f(t) = 1$ only on $-a < t < a$,

$$F(\omega) = \int_{-a}^a e^{-i\omega t} dt.$$

Integrate:

$$F(\omega) = \left[\frac{e^{-i\omega t}}{-i\omega} \right]_{-a}^a.$$

Substitute the limits:

$$F(\omega) = \frac{e^{-i\omega a} - e^{i\omega a}}{-i\omega}.$$

Using

$$e^{i\theta} - e^{-i\theta} = 2i \sin \theta,$$

we get

$$e^{-i\omega a} - e^{i\omega a} = -2i \sin(\omega a).$$

Therefore,

$$F(\omega) = \frac{-2i \sin(\omega a)}{-i\omega} = \frac{2 \sin(\omega a)}{\omega}.$$

Thus,

$$\boxed{F(\omega) = \frac{2 \sin(\omega a)}{\omega}}.$$

At $\omega = 0$, the expression has the indeterminate form $0/0$. Using the limit,

$$F(0) = \lim_{\omega \rightarrow 0} \frac{2 \sin(\omega a)}{\omega} = 2a.$$

Example

Example 2: Fourier transform of $e^{-a|t|}$, $a > 0$

Let

$$f(t) = e^{-a|t|}, \quad a > 0.$$

Because

$$|t| = \begin{cases} -t, & t < 0, \\ t, & t > 0, \end{cases}$$

we write

$$e^{-a|t|} = \begin{cases} e^{at}, & t < 0, \\ e^{-at}, & t > 0. \end{cases}$$

Using the Fourier transform definition,

$$F(\omega) = \int_{-\infty}^0 e^{at} e^{-i\omega t} dt + \int_0^{\infty} e^{-at} e^{-i\omega t} dt.$$

For the first integral,

$$\int_{-\infty}^0 e^{(a-i\omega)t} dt = \left[\frac{e^{(a-i\omega)t}}{a-i\omega} \right]_{-\infty}^0 = \frac{1}{a-i\omega}.$$

For the second integral,

$$\int_0^{\infty} e^{-(a+i\omega)t} dt = \left[\frac{e^{-(a+i\omega)t}}{-(a+i\omega)} \right]_0^{\infty} = \frac{1}{a+i\omega}.$$

Therefore,

$$F(\omega) = \frac{1}{a-i\omega} + \frac{1}{a+i\omega}.$$

Combine the fractions:

$$F(\omega) = \frac{(a+i\omega) + (a-i\omega)}{(a-i\omega)(a+i\omega)}.$$

$$F(\omega) = \frac{2a}{a^2 + \omega^2}.$$

Thus,

$$F(\omega) = \frac{2a}{\omega^2 + a^2}.$$

Example

Example 3: Modulation of a rectangular pulse

Let

$$f(t) = \begin{cases} 1, & |t| < a, \\ 0, & |t| > a. \end{cases}$$

From Example 1,

$$F(\omega) = \frac{2 \sin(\omega a)}{\omega}.$$

Now consider the modulated signal

$$g(t) = f(t) \cos(\omega_0 t).$$

Using

$$\cos(\omega_0 t) = \frac{1}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t}),$$

we have

$$g(t) = \frac{1}{2}f(t)e^{i\omega_0 t} + \frac{1}{2}f(t)e^{-i\omega_0 t}.$$

Using the frequency translation property,

$$\mathcal{F}[f(t)e^{i\omega_0 t}] = F(\omega - \omega_0),$$

and

$$\mathcal{F}[f(t)e^{-i\omega_0 t}] = F(\omega + \omega_0).$$

Thus,

$$G(\omega) = \frac{1}{2}F(\omega - \omega_0) + \frac{1}{2}F(\omega + \omega_0).$$

Substitute the rectangular pulse transform:

$$G(\omega) = \frac{\sin[(\omega - \omega_0)a]}{\omega - \omega_0} + \frac{\sin[(\omega + \omega_0)a]}{\omega + \omega_0}.$$

Engineering Insight

Engineering Note

The Fourier transform is central to signal processing. It allows engineers to identify which frequencies are present in a signal. Filtering, modulation, noise reduction, image compression, and vibration analysis all depend on understanding how signals behave in the frequency domain.

Common Mistakes

Key Concept

Common mistakes in Fourier transforms include:

1. Forgetting the factor $\frac{1}{2\pi}$ in the inverse transform.
2. Confusing ω with ordinary frequency f .
3. Assuming every function has an ordinary Fourier transform.
4. Ignoring phase and looking only at amplitude.
5. Forgetting that time shifting changes phase but not magnitude.

1.3 Hilbert Transform

Motivation and Intuition

The Hilbert transform is another linear transform used heavily in signal processing and communication theory. While the Fourier transform separates a signal into frequencies, the Hilbert transform is often used to create a phase-shifted version of a signal.

It is especially important in constructing analytic signals, where a real signal is paired with its Hilbert transform to form a complex signal.

Key Concept

The Hilbert transform of a function $x(t)$ is commonly denoted by $\hat{x}(t)$ or $H[x(t)]$ and is defined by the Cauchy principal value integral

$$\hat{x}(t) = H[x(t)] = \frac{1}{\pi} \text{P.V.} \int_{-\infty}^{\infty} \frac{x(\tau)}{t - \tau} d\tau.$$

The notation P.V. means principal value. This is necessary because the denominator becomes zero when $\tau = t$.

Physical Interpretation

The Hilbert transform shifts the phase of frequency components. In the frequency domain, it corresponds to multiplication by $-i \operatorname{sgn}(\omega)$.

Key Concept

If

$$X(\omega) = \mathcal{F}[x(t)],$$

then the Fourier transform of the Hilbert transform is

$$\mathcal{F}[\hat{x}(t)] = -i \operatorname{sgn}(\omega) X(\omega),$$

where

$$\operatorname{sgn}(\omega) = \begin{cases} 1, & \omega > 0, \\ 0, & \omega = 0, \\ -1, & \omega < 0. \end{cases}$$

Thus, positive frequency components are multiplied by $-i$, corresponding to a phase shift of $-\pi/2$, while negative frequency components are multiplied by i , corresponding to a phase shift of $\pi/2$.

Key Properties

Key Concept

Linearity

$$H[c_1x(t) + c_2y(t)] = c_1H[x(t)] + c_2H[y(t)].$$

Key Concept

Frequency-Domain Form

$$\hat{X}(\omega) = -i \operatorname{sgn}(\omega)X(\omega).$$

Key Concept

Analytic Signal The analytic signal associated with $x(t)$ is

$$z(t) = x(t) + i\hat{x}(t).$$

The real part is the original signal, and the imaginary part is its Hilbert transform.

Hilbert Transforms of Common Functions

Key Concept

Using the convention

$$\mathcal{F}[H[x(t)]] = -i \operatorname{sgn}(\omega)X(\omega),$$

the Hilbert transform shifts positive-frequency components by $-\pi/2$ and negative-frequency components by $\pi/2$.

Function $x(t)$	Hilbert Transform $H[x(t)]$
1	0
$\cos(\omega_0 t)$	$\sin(\omega_0 t)$
$\sin(\omega_0 t)$	$-\cos(\omega_0 t)$
$e^{i\omega_0 t}$	$-i \operatorname{sgn}(\omega_0)e^{i\omega_0 t}$
$\delta(t)$	$\frac{1}{\pi t}$
$\frac{1}{\pi t}$	$-\delta(t)$

Key Concept

For a real signal $x(t)$, the analytic signal is

$$z(t) = x(t) + iH[x(t)].$$

Common analytic signal examples are:

Real Signal $x(t)$	Analytic Signal $z(t)$
$\cos(\omega_0 t)$	$e^{i\omega_0 t}$
$\sin(\omega_0 t)$	$-ie^{i\omega_0 t}$

Engineering Note

The Hilbert transform is useful when the phase structure of a signal matters. In communication systems, it helps form analytic signals, amplitude envelopes, instantaneous phase, and single-sideband modulated signals.

Worked Examples

Example

Example 1: Hilbert transform of a cosine signal

Let

$$x(t) = \cos(\omega_0 t).$$

Using Euler's formula,

$$\cos(\omega_0 t) = \frac{1}{2} (e^{i\omega_0 t} + e^{-i\omega_0 t}).$$

The Fourier transform contains impulses at $\omega = \omega_0$ and $\omega = -\omega_0$:

$$X(\omega) = \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)].$$

Apply the Hilbert transform frequency rule:

$$\hat{X}(\omega) = -i \operatorname{sgn}(\omega) X(\omega).$$

At $\omega = \omega_0 > 0$,

$$-i \operatorname{sgn}(\omega_0) = -i.$$

At $\omega = -\omega_0 < 0$,

$$-i \operatorname{sgn}(-\omega_0) = i.$$

Thus,

$$\hat{X}(\omega) = -i\pi\delta(\omega - \omega_0) + i\pi\delta(\omega + \omega_0).$$

This is the Fourier transform of

$$\sin(\omega_0 t).$$

Therefore,

$$\boxed{H[\cos(\omega_0 t)] = \sin(\omega_0 t)}.$$

Example

Example 2: Hilbert transform of a sine signal

Let

$$x(t) = \sin(\omega_0 t).$$

Using Euler's formula,

$$\sin(\omega_0 t) = \frac{1}{2i} (e^{i\omega_0 t} - e^{-i\omega_0 t}).$$

Its Hilbert transform shifts the positive-frequency component by $-\pi/2$ and the negative-frequency component by $\pi/2$.

Using the known phase-shift relation,

$$H[\sin(\omega_0 t)] = -\cos(\omega_0 t).$$

Therefore,

$$\boxed{H[\sin(\omega_0 t)] = -\cos(\omega_0 t)}.$$

Example

Example 3: Analytic signal for $x(t) = \cos(\omega_0 t)$

From Example 1,

$$H[\cos(\omega_0 t)] = \sin(\omega_0 t).$$

The analytic signal is

$$z(t) = x(t) + i\hat{x}(t).$$

Substitute:

$$z(t) = \cos(\omega_0 t) + i\sin(\omega_0 t).$$

Using Euler's formula,

$$z(t) = e^{i\omega_0 t}.$$

Thus, the analytic signal associated with $\cos(\omega_0 t)$ is

$$z(t) = e^{i\omega_0 t}.$$

Engineering Insight

Engineering Note

The Hilbert transform is used in communication systems to form analytic signals, determine instantaneous phase, compute instantaneous frequency, and analyze amplitude envelopes. It is also used in single-sideband modulation, radar signal processing, and spectral-response analysis.

Common Mistakes

Key Concept

Common mistakes in Hilbert transforms include:

1. Ignoring the principal value nature of the integral.
2. Treating the Hilbert transform as an ordinary time delay.
3. Forgetting that it changes phase but not amplitude.
4. Confusing the analytic signal $z(t)$ with the original real signal $x(t)$.
5. Applying the transform without considering whether the signal is suitable.